

Qualification Pack



Casting Operator - Metalware

QP Code: HCS/Q2801

Version: 4.0

NSQF Level: 4

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HCS/Q2801: Casting Operator - Metalware

Brief Job Description

A metalware casting operator is responsible for the precise operation of casting processes to produce high-quality metal components. They ensure the accuracy of moulds and oversee the casting process, maintaining safety and efficiency throughout production.

Personal Attributes

A metalware casting operator is detail-oriented and safety-conscious, demonstrating strong problem-solving skills and a commitment to quality. They are adaptable and efficient, with a solid understanding of casting techniques and processes.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HCS/N2801: Perform pre casting requirements](#)
2. [HCS/N2802: Casting of metal surface](#)
3. [HCS/N0603: Carry out the process involved in Sandcasting Techniques](#)
4. [HCS/N8505: Entrepreneurship skills for Metal casting operator](#)
5. [HCS/N9901: Coordinate with colleagues and work as a team](#)
6. [HCS/N9913: Maintain health, safety and security at workplace](#)
7. [HCS/N9903: Maintain Personal Hygiene](#)
8. [HCS/N9904: Basic Business Management](#)
9. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Handicrafts and Carpet
Sub-Sector	Metalware
Occupation	Metal Casting and Stamping

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Country	India
NSQF Level	4
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7211.0100
Minimum Educational Qualification & Experience	12th grade Pass with NA of experience OR 8th grade pass with 1.5 years of experience OR 5th grade pass with 3 Years of experience OR Ability to read and write with 5 Years of experience OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	4.0
Reference code on NQR	QG-04-HC-03850-2025-V2-HCSSC
NQR Version	4

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HCS/N2801: Perform pre casting requirements

Description

This OS unit is about preparing the pre casting requirements to cast the metal surface

Scope

The scope covers the following :

- understand work requirement
- assemble the required raw materials
- set the cast machine

Elements and Performance Criteria

Understand work requirement

To be competent, the user/individual on the job must be able to:

- PC1.** Receive work instruction from the supervisor.
- PC2.** Understand the job sheet and the casting method.
- PC3.** Plan the target on the number of pieces to be completed.

Assemble the required raw materials

To be competent, the user/individual on the job must be able to:

- PC4.** Select the appropriate tool and machine for casting such as coal, coarse blade, crucible, furnace, refrigerator bricks.
- PC5.** Arrange for the furnace preparation.
- PC6.** Collect and arrange the materials to begin the process.
- PC7.** Report on any shortage or defect of raw materials to the concerned person.
- PC8.** Ensure to stock the required materials in advance.
- PC9.** Collect the metal product to be casted.

Set the cast machine

To be competent, the user/individual on the job must be able to:

- PC10.** Check for the working condition of the casting machine. in case, if using the machine.
- PC11.** Understand the operation of the machine for dos and don'ts.
- PC12.** Set the machine to begin the process.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company's policies on: incentives, safety and hazards, personnel management and quality standards
- KU2.** The workflow involved in metal making process of the company
- KU3.** Importance of the individual's role in the work process

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- KU4.** Documentation policy
- KU5.** Metal making process and types of products
- KU6.** Various kinds of raw materials involved in the process
- KU7.** Different kinds of casting methods to be used based on the requirement
- KU8.** Furnace preparation and operation
- KU9.** Appropriate temperature to heat
- KU10.** Tools, consumables and equipment/types of equipment for various kinds of casting
- KU11.** Process adopted for various casting methods such as sand casting, mold casting, etc.
- KU12.** Handling the tools and equipment/types of equipment for casting
- KU13.** Maintaining the tools and equipment/types of equipment for casting
- KU14.** Safety standards and precautions to be taken
- KU15.** Quality standards to be maintained

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** take notes or read about metal making techniques
- GS2.** read and write the notes from the design
- GS3.** read company policy documents
- GS4.** interact with team members to work efficiently
- GS5.** communicate effectively with the supervisor
- GS6.** share workload with the colleagues in the process
- GS7.** multi task and deliver the final finished piece on time adhering to quality standards
- GS8.** plan for daily production
- GS9.** plan for budget and material requirement
- GS10.** customer preference, taste, etc and accordingly make crafts
- GS11.** solve issues relating to material, cost and labour and ensure smooth production
- GS12.** analyse the material requirement, corrective action required during casting
- GS13.** spot process disruptions and delays
- GS14.** use the various casting tools such as coal, coarse blade, crucible, furnace, refrigerator bricks
- GS15.** use the cast machine

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand work requirement</i>	7	9	-	-
PC1. Receive work instruction from the supervisor.	2	3	-	-
PC2. Understand the job sheet and the casting method.	2	3	-	-
PC3. Plan the target on the number of pieces to be completed.	3	3	-	-
<i>Assemble the required raw materials</i>	18	36	-	-
PC4. Select the appropriate tool and machine for casting such as coal, coarse blade, crucible, furnace, refrigerator bricks.	3	6	-	-
PC5. Arrange for the furnace preparation.	3	6	-	-
PC6. Collect and arrange the materials to begin the process.	3	6	-	-
PC7. Report on any shortage or defect of raw materials to the concerned person.	3	6	-	-
PC8. Ensure to stock the required materials in advance.	3	6	-	-
PC9. Collect the metal product to be casted.	3	6	-	-
<i>Set the cast machine</i>	12	18	-	-
PC10. Check for the working condition of the casting machine. in case, if using the machine.	4	6	-	-
PC11. Understand the operation of the machine for dos and don'ts.	4	6	-	-
PC12. Set the machine to begin the process.	4	6	-	-
NOS Total	37	63	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HCS/N2801
NOS Name	Perform pre casting requirements
Sector	Handicrafts and Carpet
Sub-Sector	Metalware
Occupation	Metal Casting and Stamping
NSQF Level	4
Credits	2
Version	2.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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HCS/N2802: Casting of metal surface

Description

This OS unit is about preparing the mold, melting metal, pouring the liquid metal into a mold, solidify and cool to attain the desired shape as per the requirement.

Scope

The scope covers the following :

- prepare the mold
- operate the furnace and melt the metal
- pour the molten metal
- clean the casted metal
- check for defects
- undertake preventive maintenance of the cast machine
- achieve productivity and quality standards

Elements and Performance Criteria

prepare the mold

To be competent, the user/individual on the job must be able to:

- PC1.** decide on the type of mold to be prepared, if closed mold or open mold
- PC2.** ensure to use the appropriate tools for molding
- PC3.** clean the surface before beginning to mold
- PC4.** prepare the pattern in the form of a model using the appropriate materials
- PC5.** prepare the mold by holding around a pattern within a supporting metal frame
- PC6.** remove the pattern leaving the mold cavity with cores
- PC7.** ensure the mold is prepared to the required shape

operate the furnace and melt the metal

To be competent, the user/individual on the job must be able to:

- PC8.** decide on the type of casting to be performed from expendable, permanent mold casting, sand casting, investment casting, etc.
- PC9.** prepare the molten metal melting a solid metal
- PC10.** determine the size of the furnace
- PC11.** heat this solid metal placed in coal to an appropriate temperature in a suitable furnace for required time
- PC12.** monitor the furnace constantly to avoid defects at later stage
- PC13.** open the furnace and check for completion

pour the molten metal

To be competent, the user/individual on the job must be able to:

- PC14.** pour the molten metal into the mold at the right temperature manually
- PC15.** ensure not to leak while pouring

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- PC16.** ensure minimal accidents while handling molten metal
- PC17.** allow this to solidify and cool to room temperature
- PC18.** take the metal out of the mold, trim and bring to the desired shape
- PC19.** ensure to set the appropriate temperature and duration setting to the machine for operation as per specifications
- PC20.** ensure to weigh the correct amount of lubricants to be used

clean the casted metal

To be competent, the user/individual on the job must be able to:

- PC21.** clean the surface of the cast and improve its appearance

check for defects

To be competent, the user/individual on the job must be able to:

- PC22.** check the metal after casting, if it matches requirement
- PC23.** make necessary adjustments if any
- PC24.** rectify and rework if any mistakes are found un-matching the design specifications and requirements

- PC25.** melt and recast product returned from qc

undertake preventive maintenance of the cast machine

To be competent, the user/individual on the job must be able to:

- PC26.** ensure general maintenance of the machine
- PC27.** ensure no shut down of machines due to improper maintenance

achieve productivity and quality standards

To be competent, the user/individual on the job must be able to:

- PC28.** ensure casting process performed is as per requirement
- PC29.** ensure the target number of pieces are casted
- PC30.** ensure the output delivered is defect free and hazard free

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's policies on: incentives, safety and hazards, personnel management and quality standards
- KU2.** workflow involved in metal making process of the company
- KU3.** importance of the individual's role in the work process
- KU4.** reporting structure
- KU5.** documentation policy
- KU6.** customer profile
- KU7.** metal craft details
- KU8.** metal making process and types of products
- KU9.** various kinds of raw materials involved in the process
- KU10.** different kinds of casting methods to be used based on the requirement
- KU11.** cast machine use and maintenance

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- KU12.** furnace preparation and operation
- KU13.** appropriate temperature to heat
- KU14.** preparation of mold
- KU15.** melting and careful pouring of liquid metal
- KU16.** cleaning requirements for casting metal
- KU17.** preparation of the metal surface
- KU18.** work hazards involved in the cast process
- KU19.** tools, consumables and equipment/types of equipment for various kinds of casting
- KU20.** process adopted for various casting methods such as sand casting, mold casting, etc.
- KU21.** bringing the required tools and equipment/types of equipment to the desired shape
- KU22.** handling the tools and equipment/types of equipment for casting
- KU23.** maintaining the tools and equipment/types of equipment for casting
- KU24.** creating tools and equipment/types of equipment for casting
- KU25.** use of hazardous acids and chemicals
- KU26.** safety standards and precautions to be taken
- KU27.** quality standards to be maintained
- KU28.** standard operating procedure
- KU29.** market trend and customer preferences

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** take notes or read about metal making techniques
- GS2.** read and write the notes from the design
- GS3.** read company policy documents
- GS4.** interact with team members to work efficiently
- GS5.** communicate effectively with the supervisor
- GS6.** share work load with the colleagues in the process
- GS7.** multi task and deliver the final finished piece on time adhering to quality standards
- GS8.** plan for daily production
- GS9.** plan for budget and material requirement
- GS10.** customer preference, taste, etc and accordingly make crafts
- GS11.** solve issues relating to material, cost and labour and ensure smooth production
- GS12.** analyse the material requirement, corrective action required during casting
- GS13.** spot process disruptions and delays
- GS14.** use the various casting tools such as coal, coarse blade, crucible, furnace, refrigerator bricks
- GS15.** use the cast machine

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>prepare the mold</i>	7	16	-	-
PC1. decide on the type of mold to be prepared, if closed mold or open mold	1	2	-	-
PC2. ensure to use the appropriate tools for molding	1	2	-	-
PC3. clean the surface before beginning to mold	1	2	-	-
PC4. prepare the pattern in the form of a model using the appropriate materials	1	2	-	-
PC5. prepare the mold by holding around a pattern within a supporting metal frame	1	3	-	-
PC6. remove the pattern leaving the mold cavity with cores	1	3	-	-
PC7. ensure the mold is prepared to the required shape	1	2	-	-
<i>operate the furnace and melt the metal</i>	6	13	-	-
PC8. decide on the type of casting to be performed from expendable, permanent mold casting, sand casting, investment casting, etc.	1	2	-	-
PC9. prepare the molten metal melting a solid metal	1	2	-	-
PC10. determine the size of the furnace	1	2	-	-
PC11. heat this solid metal placed in coal to an appropriate temperature in a suitable furnace for required time	1	3	-	-
PC12. monitor the furnace constantly to avoid defects at later stage	1	2	-	-
PC13. open the furnace and check for completion	1	2	-	-
<i>pour the molten metal</i>	7	15	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. pour the molten metal into the mold at the right temperature manually	1	3	-	-
PC15. ensure not to leak while pouring	1	2	-	-
PC16. ensure minimal accidents while handling molten metal	1	2	-	-
PC17. allow this to solidify and cool to room temperature	1	2	-	-
PC18. take the metal out of the mold, trim and bring to the desired shape	1	2	-	-
PC19. ensure to set the appropriate temperature and duration setting to the machine for operation as per specifications	1	2	-	-
PC20. ensure to weigh the correct amount of lubricants to be used	1	2	-	-
<i>clean the casted metal</i>	1	2	-	-
PC21. clean the surface of the cast and improve its appearance	1	2	-	-
<i>check for defects</i>	4	8	-	-
PC22. check the metal after casting, if it matches requirement	1	2	-	-
PC23. make necessary adjustments if any	1	2	-	-
PC24. rectify and rework if any mistakes are found un-matching the design specifications and requirements	1	2	-	-
PC25. melt and recast product returned from qc	1	2	-	-
<i>undertake preventive maintenance of the cast machine</i>	2	4	-	-
PC26. ensure general maintenance of the machine	1	2	-	-
PC27. ensure no shut down of machines due to improper maintenance	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>achieve productivity and quality standards</i>	6	9	-	-
PC28. ensure casting process performed is as per requirement	2	3	-	-
PC29. ensure the target number of pieces are casted	2	3	-	-
PC30. ensure the output delivered is defect free and hazard free	2	3	-	-
NOS Total	33	67	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HCS/N2802
NOS Name	Casting of metal surface
Sector	Handicrafts and Carpet
Sub-Sector	Metalware
Occupation	Metal Casting and Stamping
NSQF Level	4
Credits	3
Version	2.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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HCS/N0603: Carry out the process involved in Sandcasting Techniques

Description

This unit focuses on the Sandcasting is a metal casting process where molten metal is poured into a sand mold to create objects. First, a pattern of the object is made and placed in a two-part mold filled with sand. Once the pattern is removed, the mold is reassembled, and molten metal is poured into the cavity. After the metal cools and hardens, the mold is broken, and the final metal part is cleaned and finished.

Scope

The scope covers the following :

- design the pattern
- mold preparation
- pouring the molten metal
- cooling and removing the casting
- finishing touches

Elements and Performance Criteria

Design the Pattern

To be competent, the user/individual on the job must be able to:

- PC1.** create a pattern of the object to be cast
- PC2.** model or pattern of the desired part is made from wood, metal, or plastic

Mold Preparation

To be competent, the user/individual on the job must be able to:

- PC3.** Place a pattern inside a two part frame called a "flask"
- PC4.** mix casting sand with clay and water and tightly packed around the pattern

Pouring the Molten Metal

To be competent, the user/individual on the job must be able to:

- PC5.** heat the metal in a furnace until it becomes melt
- PC6.** close the mold and pour the molten metal through the gating system into the cavity
- PC7.** fills the cavity and takes the shape of the part

Cooling and Removing the Casting

To be competent, the user/individual on the job must be able to:

- PC8.** allow the metal cool and solidify inside the sand mold
- PC9.** clean the casting and remove the excess sand or any unwanted metal (like the sprue and gates) is cut off

Finishing Touches

To be competent, the user/individual on the job must be able to:

- PC10.** polish and sand the casting to remove rough surfaces, imperfections, or sharp edges
- PC11.** trim the excess metal or flash (the thin layer of metal along the mold lines) by using cutting tools

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- PC12.** apply finishing process such as grinding, polishing, or coating, depending on the desired surface finish

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understand the silica sand is commonly used due to its excellent thermal properties and ability to hold the shape
- KU2.** various kinds of raw materials involved in the process
- KU3.** various metals, including aluminum, bronze, and iron, can be cast using this technique
- KU4.** knowledge of a pattern (the replica of the final product) is created, often made from wood, metal, or plastic
- KU5.** understand the halves of the mold (cope and drag) are assembled
- KU6.** handling the tools and equipment/types of equipment for sand casting
- KU7.** furnace preparation and operation
- KU8.** appropriate temperature to heat
- KU9.** quality standards to be maintained
- KU10.** conduct inspection and testing process to ensure the final product meets specifications

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** take notes or read about sand casting techniques
- GS2.** read and write the notes from the designing pattern
- GS3.** interact with team members to work efficiently
- GS4.** communicate effectively with the supervisor
- GS5.** customer preference, taste, etc and accordingly make crafts
- GS6.** solve issues relating to material, cost and labour and ensure smooth production
- GS7.** skills in assembling and preparing the mold, including ensuring proper venting and gating systems
- GS8.** identify common defects (e.g., porosity, shrinkage) and understand their causes and solutions

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design the Pattern</i>	4	11	-	-
PC1. create a pattern of the object to be cast	2	6	-	-
PC2. model or pattern of the desired part is made from wood, metal, or plastic	2	5	-	-
<i>Mold Preparation</i>	5	12	-	-
PC3. Place a pattern inside a two part frame called a "flask"	2	6	-	-
PC4. mix casting sand with clay and water and tightly packed around the pattern	3	6	-	-
<i>Pouring the Molten Metal</i>	8	18	-	-
PC5. heat the metal in a furnace until it becomes melt	3	6	-	-
PC6. close the mold and pour the molten metal through the gating system into the cavity	2	6	-	-
PC7. fills the cavity and takes the shape of the part	3	6	-	-
<i>Cooling and Removing the Casting</i>	6	11	-	-
PC8. allow the metal cool and solidfy inside the sand mold	3	5	-	-
PC9. clean the casting and remove the excess sand or any unwanted metal (like the sprue and gates) is cut off	3	6	-	-
<i>Finishing Touches</i>	7	18	-	-
PC10. polish and sand the casting to remove rough surfaces, imperfections, or sharp edges	3	6	-	-
PC11. trim the excess metal or flash (the thin layer of metal along the mold lines) by using cutting tools	2	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. apply finishing process such as grinding, polishing, or coating, depending on the desired surface finish	2	6	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0603
NOS Name	Carry out the process involved in Sandcasting Techniques
Sector	Handicrafts and Carpet
Sub-Sector	
Occupation	Casting
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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HCS/N8505: Entrepreneurship skills for Metal casting operator

Description

In this unit focuses on the casting sector of metal handicrafts involves creating intricate designs through processes like sand casting. Entrepreneurs in this field must possess a blend of technical skills, such as understanding casting techniques and material properties, along with creative design abilities to craft appealing products

Scope

The scope covers the following :

- technical skills
- business management skills
- creativity and innovation
- networking and relationship building
- operational skills
- sustainability awareness

Elements and Performance Criteria

Technical Skills

To be competent, the user/individual on the job must be able to:

- PC1.** learn about various types of casting techniques
- PC2.** identify the different metal and their properties helps in selecting the appropriate material for specific designs and ensuring quality

Business Management Skills

To be competent, the user/individual on the job must be able to:

- PC3.** analyze the market trends, consumer preferences, and competitor offerings helps in identifying potential opportunities and niches
- PC4.** use supply chain dynamics, including sourcing raw materials and managing logistics, is crucial for smooth operations

Creativity and Innovation

To be competent, the user/individual on the job must be able to:

- PC5.** ability to innovate and develop new products that meet consumer needs, utilizing unique designs or materials
- PC6.** create new product designs or improve existing ones to stand out in a competitive market
- PC7.** use creative thinking to address production challenges, quality issues, and market demands
- PC8.** Keep up with current design trends and consumer preferences can help entrepreneurs develop products that resonate with target audiences

Operational Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use production workflows and optimizing processes to improve efficiency and reduce waste is essential for scaling operations

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PC10. identify the safety regulations and practices in metal casting is vital to ensure a safe working environment for employees

Sustainability Awareness

To be competent, the user/individual on the job must be able to:

PC11. identify the sustainable practices and materials can help in developing environmentally friendly products, appealing to eco-conscious consumers

PC12. implement the effective waste management practices in the casting process is essential for reducing environmental impact and improving efficiency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. familiarity with various casting techniques is essential to determine the most suitable method for specific products

KU2. properties of different metals (e.g., aluminum, brass, bronze) and alloys helps in selecting the right material for durability and aesthetic appeal

KU3. machinery and tools involved in the casting process, including furnaces, molds, and finishing tools, is crucial for effective production

KU4. aware of the current trends in metal handicrafts and understanding consumer preferences can help in product development and marketing strategies

KU5. analyze the competitors in the market to identify unique selling propositions and gaps that can be filled with new products

KU6. importance of sustainable practices in the casting sector, including using recyclable materials and energy-efficient production methods

KU7. awareness of environmental regulations and standards that impact metal casting operations

KU8. significance of metal handicrafts and traditional techniques

KU9. recognize how heritage and cultural aspects can be leveraged in branding and marketing strategies

KU10. build relationships with suppliers, artisans, designers, and customers can lead to collaborative opportunities and strengthen business networks

KU11. importance of engaging with local communities and artisans can enhance brand authenticity and social responsibility

KU12. emphasize the mindset of continuous improvement and innovation in design, processes, and customer service to stay competitive

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. learn the standard procedure for metal casting processes

GS2. write notes on mold-making techniques and materials used in casting

GS3. read and write the notes on different metals and their properties

GS4. quality standards in metal casting

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- GS5.** inspecting and testing cast products to ensure they meet specifications
- GS6.** assess market demand and adjust production accordingly
- GS7.** effective marketing strategies to promote metal handicrafts
- GS8.** importance of fostering a positive workplace culture
- GS9.** embrace new technologies and methods in the casting process
- GS10.** work with other artisans, designers, and businesses to create synergies and enhance product offerings

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Technical Skills</i>	7	10	-	-
PC1. learn about various types of casting techniques	3	5	-	-
PC2. identify the different metal and their properties helps in selecting the appropriate material for specific designs and ensuring quality	4	5	-	-
<i>Business Management Skills</i>	6	10	-	-
PC3. analyze the market trends, consumer preferences, and competitor offerings helps in identifying potential opportunities and niches	3	5	-	-
PC4. use supply chain dynamics, including sourcing raw materials and managing logistics, is crucial for smooth operations	3	5	-	-
<i>Creativity and Innovation</i>	14	20	-	-
PC5. ability to innovate and develop new products that meet consumer needs, utilizing unique designs or materials	3	5	-	-
PC6. create new product designs or improve existing ones to stand out in a competitive market	4	5	-	-
PC7. use creative thinking to address production challenges, quality issues, and market demands	4	5	-	-
PC8. Keep up with current design trends and consumer preferences can help entrepreneurs develop products that resonate with target audiences	3	5	-	-
<i>Operational Skills</i>	7	10	-	-
PC9. use production workflows and optimizing processes to improve efficiency and reduce waste is essential for scaling operations	4	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. identify the safety regulations and practices in metal casting is vital to ensure a safe working environment for employees	3	5	-	-
<i>Sustainability Awareness</i>	6	10	-	-
PC11. identify the sustainable practices and materials can help in developing environmentally friendly products, appealing to eco-conscious consumers	3	5	-	-
PC12. implement the effective waste management practices in the casting process is essential for reducing environmental impact and improving efficiency	3	5	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HCS/N8505
NOS Name	Entrepreneurship skills for Metal casting operator
Sector	Handicrafts and Carpet
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

HCS/N9901: Coordinate with colleagues and work as a team

Description

This NOS describes the communication and coordination skills and knowledge to work with colleagues and supervisor to achieve a smooth and hazard-free workflow.

Scope

The scope covers the following :

- interact with supervisor or superior
- work as a team by coordinating with colleagues within and outside the department and include inputs on PwD & Gender Sensitisation
- report and Document

Elements and Performance Criteria

Interact with supervisor or superior

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health, safety gender, and PwD (People with disability) related instructions applicable to the workplace
- PC2.** actively participate in mock drills/ evacuation procedures; group discussions, training sensitization programs for gender, and PwD awareness organized at the workplace.
- PC3.** receive job orders and instructions from reporting supervisor and receive feedback on work standards.
- PC4.** understand the work output requirements, targets, performance indicators and incentives.
- PC5.** deliver quality work on time and report any anticipated reasons for delays and handover completed work to supervisor
- PC6.** report on any grievances, production defects and any potential hazards.

Work as a team by coordinating with colleagues within and outside the department and include inputs on PwD & Gender Sensitisation

To be competent, the user/individual on the job must be able to:

- PC7.** communicate maintenance and repair schedule proactively to the supervisor
- PC8.** interact and clarify doubts on design, usage of materials & tools, quality & standards compliance, etc.

Report and Document

To be competent, the user/individual on the job must be able to:

- PC9.** report in time for shortage or need of raw materials
- PC10.** communicate with colleagues from within and other departments, clearly and effectively on all aspects to carry out the work among the team
- PC11.** maintain the etiquette, use polite language, demonstrate responsible and disciplined behavior towards colleagues.
- PC12.** put team over individual goals and multi-task or share work where necessary supporting the colleagues.

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PC13. document all the details accurately relating to ones role as required.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** knowledge about the importance of gender equality being followed in the organization and policies for reporting any harassment or inappropriate behavior
- KU2.** knowledge about how to accommodate employees with disabilities etiquette to adhere to and proper language and terminology
- KU3.** knowledge about how to communicate, offer help, respecting space, parking etc. for people with disabilities or special needs
- KU4.** knowledge about promoting a safe, accessible and healthy workplace for disabled employees
- KU5.** company's policies on preferred language of communication, incentives, quality standards, personnel management, reporting and escalation matrix policy.
- KU6.** company's standard operating procedure (sop) and the risk and impact of not following them.
- KU7.** procedures for working with colleagues, his/her role and responsibilities in relation to this
- KU8.** organizational hierarchy and the line of reporting structure and work target and review mechanism
- KU9.** procedures to report employment related issues and to deal with conflicts
- KU10.** importance of the individuals role in the organizational workflow and details of the individual responsibilities
- KU11.** tools and equipment handling procedure and common potential hazards in the work place and the procedures to deal with them
- KU12.** effective communication with various categories of people and the different departments in the organization
- KU13.** to document the job activity as required like the check sheets, history sheets, etc
- KU14.** expressing and addressing grievances appropriately, deal with difficult work relationships and manage the internal conflicts effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** create a positive and inclusive workplace atmosphere without any kind of bias/discrimination towards any employee
- GS2.** actively take part in any discussion/workshop organized for gender sensitization training
- GS3.** read and comprehend written instructions related to gender equality issues in the organization
- GS4.** identify and report any harassment or inappropriate behavior towards any employee
- GS5.** create a positive and inclusive workplace atmosphere without any kind of bias/discrimination towards any employee with disability or special needs
- GS6.** actively take part in any discussion/workshop organized for disability sensitization training.

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- GS7.** read and comprehend written instructions related to equality issues in the organization related to disabled persons
- GS8.** read job sheets, design sheet and information displayed at the workplace
- GS9.** read and understand manuals, health and safety instructions, memos etc
- GS10.** fill up documentation to ones role
- GS11.** communicate effectively with supervisor
- GS12.** contribute to quality of team work and achieve smooth workflow
- GS13.** improve work processes by interacting with others and adopting best practices

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interact with supervisor or superior</i>	14	30	-	-
PC1. comply with health, safety gender, and PwD (People with disability) related instructions applicable to the workplace	2	5	-	-
PC2. actively participate in mock drills/ evacuation procedures; group discussions, training sensitization programs for gender, and PwD awareness organized at the workplace.	2	5	-	-
PC3. receive job orders and instructions from reporting supervisor and receive feedback on work standards.	2	5	-	-
PC4. understand the work output requirements, targets, performance indicators and incentives.	2	5	-	-
PC5. deliver quality work on time and report any anticipated reasons for delays and handover completed work to supervisor	3	5	-	-
PC6. report on any grievances, production defects and any potential hazards.	3	5	-	-
<i>Work as a team by coordinating with colleagues within and outside the department and include inputs on PwD & Gender Sensitisation</i>	6	10	-	-
PC7. communicate maintenance and repair schedule proactively to the supervisor	3	5	-	-
PC8. interact and clarify doubts on design, usage of materials & tools, quality & standards compliance, etc.	3	5	-	-
<i>Report and Document</i>	15	25	-	-
PC9. report in time for shortage or need of raw materials	3	5	-	-
PC10. communicate with colleagues from within and other departments, clearly and effectively on all aspects to carry out the work among the team	3	5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. maintain the etiquette, use polite language, demonstrate responsible and disciplined behavior towards colleagues.	3	5	-	-
PC12. put team over individual goals and multi-task or share work where necessary supporting the colleagues.	3	5	-	-
PC13. document all the details accurately relating to ones role as required.	3	5	-	-
NOS Total	35	65	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N9901
NOS Name	Coordinate with colleagues and work as a team
Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Ceramics), Handicrafts (Fashion Jewellery), Handicrafts (Stonecraft), Glassware, Metalware, Carpet, Handicrafts (Woodware)
Occupation	Generic
NSQF Level	4
Credits	1
Version	6.0
Last Reviewed Date	11/12/2025
Next Review Date	11/12/2028
NSQC Clearance Date	11/12/2025

Qualification Pack

HCS/N9913: Maintain health, safety and security at workplace

Description

This unit provides Performance Criteria, Knowledge & Understanding and Skills & Abilities required to comply with health, safety and security requirements at the workplace and covers procedures to prevent, control and minimise risk to self and others.

Elements and Performance Criteria

Comply with health, safety and security requirements at work

To be competent, the user/individual on the job must be able to:

- PC1.** Comply with health and safety related instructions applicable to the workplace.
- PC2.** Use and maintain personal protective equipment as per protocol.
- PC3.** Carry out own activities in line with approved guidelines and procedures.
- PC4.** Maintain a healthy lifestyle and guard against dependency on intoxicants.
- PC5.** Follow environment management system related procedures.
- PC6.** Store materials and tools in line with manufacturers and organisational requirements.
- PC7.** Safely handle and move waste and debris.
- PC8.** Minimize health and safety risks to self and others due to own actions.
- PC9.** Seek clarifications, from supervisors or other authorized personnel in case of perceived risks.
- PC10.** Monitor the workplace and work processes for potential risks and threats.
- PC11.** Carry out periodic walk-through to keep work area free from hazards and obstructions, if assigned.
- PC12.** Report hazards and potential risks/ threats to supervisors or other authorized personnel.
- PC13.** Participate in mock drills/ evacuation procedures organized at the workplace.
- PC14.** Undertake first aid, fire-fighting and emergency response training, if asked to do so.
- PC15.** Take action based on instructions in the event of fire, emergencies or accidents.
- PC16.** Follow organisation procedures for evacuation when required.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Health and safety related practices applicable at the workplace.
- KU2.** Potential hazards, risks and threats based on nature of operations.
- KU3.** Organizational procedures for safe handling of tools.
- KU4.** Potential risks due to own actions and methods to minimize these.
- KU5.** Environmental management system related procedures at the workplace.
- KU6.** Layout of the plant and details of emergency exits, escape routes, emergency equipment and assembly points.
- KU7.** Potential accidents and emergencies and response to these scenarios.

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- KU8.** Reporting protocol and documentation required.
- KU9.** Details of personnel trained in first aid, fire-fighting and emergency response.
- KU10.** Actions to take in the event of a mock drills/ evacuation procedures or actual accident, emergency or fire.
- KU11.** Occupational health and safety risks and methods.
- KU12.** Personal protective equipment and method of use.
- KU13.** Identification, handling and storage of hazardous substances.
- KU14.** Proper disposal system for waste and by-products.
- KU15.** Signage related to health and safety and their meaning.
- KU16.** Importance of sound health, hygiene and good habits.
- KU17.** Ill-effects of alcohol, tobacco and drugs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write in local language.
- GS2.** Read measurement instructions.
- GS3.** Communicate orally with colleagues.
- GS4.** Follow organization rule-based decision making process.
- GS5.** Take decision with systematic course of actions and/or response.
- GS6.** Plan and organize your work to achieve targets and deadlines.
- GS7.** Manage relationships with customers.
- GS8.** Build customer relationships and use customer centric approach.
- GS9.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s).
- GS10.** Identify immediate or temporary solutions to resolve delays.
- GS11.** Analyze data and activities.
- GS12.** Pass on relevant information to others.
- GS13.** Apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Comply with health, safety and security requirements at work</i>	30	70	-	-
PC1. Comply with health and safety related instructions applicable to the workplace.	2	6	-	-
PC2. Use and maintain personal protective equipment as per protocol.	2	6	-	-
PC3. Carry out own activities in line with approved guidelines and procedures.	2	6	-	-
PC4. Maintain a healthy lifestyle and guard against dependency on intoxicants.	2	6	-	-
PC5. Follow environment management system related procedures.	2	4	-	-
PC6. Store materials and tools in line with manufacturers and organisational requirements.	2	3	-	-
PC7. Safely handle and move waste and debris.	1	3	-	-
PC8. Minimize health and safety risks to self and others due to own actions.	2	4	-	-
PC9. Seek clarifications, from supervisors or other authorized personnel in case of perceived risks.	1	3	-	-
PC10. Monitor the workplace and work processes for potential risks and threats.	1	3	-	-
PC11. Carry out periodic walk-through to keep work area free from hazards and obstructions, if assigned.	2	3	-	-
PC12. Report hazards and potential risks/ threats to supervisors or other authorized personnel.	3	4	-	-
PC13. Participate in mock drills/ evacuation procedures organized at the workplace.	2	3	-	-
PC14. Undertake first aid, fire-fighting and emergency response training, if asked to do so.	2	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Take action based on instructions in the event of fire, emergencies or accidents.	2	6	-	-
PC16. Follow organisation procedures for evacuation when required.	2	6	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N9913
NOS Name	Maintain health, safety and security at workplace
Sector	Handicrafts and Carpet
Sub-Sector	Hand Crafted Textiles, Hand Crocheted Textiles, Handicrafts (Agarbatti), Handicrafts (Bamboo)
Occupation	Traditional Hand Embroiderer, Master Hand Embroiderer, Embroidery finishing, Design Tracer, Hand Block Printer, Block Print Supervisor, Hand Embroidery, Crochet Lace Maker, Master Crochet Lace Maker, Crochet Lace Tailor, Handmade Bamboo Agarbatti Stick Maker, Semi-Mechanized Bamboo Stick Maker, Automatic Stick Making M/C Operator, Hand Rolled Agarbatti Maker, Pedal Operated Machine Agarbatti Maker, Automatic Machine Rolled Agarbatti Maker, Agarbatti Perfume Applicator, Bamboo Processor and Dyer, Bamboo Mat Weaver, Handmade Bamboo Stick Maker (for Woven Mats), Bamboo Basket Maker, Bamboo Utility Handicraft Assembler, Bamboo Utility Product Tailor, Bamboo Artwork Maker.
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N9903: Maintain Personal Hygiene

Description

This NOS provides the abilities required for taking responsibility for their own health at the workplace and is about using the correct procedures to prevent, control and minimize risk to them and others at the workplace.

Scope

The scope covers the following :

- adopt healthy work practices
- achieve work productivity while maintaining health

Elements and Performance Criteria

Adopt healthy work practices

To be competent, the user/individual on the job must be able to:

- PC1.** always cover the mouth and nose with a dust mask while working and keep on changing when it gets blocked with dust
- PC2.** wear safety shoes while visiting the production unit to avoid any damage
- PC3.** wear personal protective equipment while visiting the different departments during production. orexample mask in the washing section, glasses and mask in an assembly line, and gloves in the printing section, etc.
- PC4.** always wash sanitize your hands after a factory unit before touching any document, laptop, cell phone, etc.
- PC5.** undergo preventive health checkups at regular intervals.
- PC6.** take prompt treatment from the doctor in case of illness.

Achieve work productivity while maintaining health

To be competent, the user/individual on the job must be able to:

- PC7.** follow SOPs for dealing with blisters; scratches; accidental fires or any other type of emergencies at work
- PC8.** ensure no productivity loss or absenteeism from work due to illness
- PC9.** ensure no long-term ill effect on personal health.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's policies on personal health and occupational hazard management
- KU2.** company's HR policies.
- KU3.** company's reporting structure
- KU4.** company's emergency evacuation procedure
- KU5.** health risks to the worker at the workplace

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- KU6.** healthy work practices
- KU7.** how to perform the duties in a way to minimize pollution at the workplace.
- KU8.** what personal protective equipment should be worn and how it is cared for
- KU9.** safe disposal methods for waste
- KU10.** how to provide first-aid treatment at the workplace
- KU11.** emergency procedures to be followed in case of an mishap such as fire accidents etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read personal health instructions and manual
- GS2.** read the usage of various hand tools and personal protection equipment
- GS3.** take notes on descriptions and details of various personal health maintenance procedures
- GS4.** communicate supervisor about the physical symptoms
- GS5.** receive instructions from doctor and supervisor on medical care
- GS6.** how to select appropriate hand tools and personal protection equipment
- GS7.** when to change personal protection equipment during work
- GS8.** how to identify first aid needs in case and of an injury
- GS9.** how to select appropriate hand tools and personal protection equipment
- GS10.** when to change personal protection equipment during work
- GS11.** how to use materials that does not affect customer health / make injury
- GS12.** improve work processes by adopting best safety practices
- GS13.** analyze the usage of appropriate tools and consumables
- GS14.** spot errors and any other disruptions and communicate with solutions

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Adopt healthy work practices</i>	18	48	-	-
PC1. always cover the mouth and nose with a dust mask while working and keep on changing when it gets blocked with dust	3	8	-	-
PC2. wear safety shoes while visiting the production unit to avoid any damage	3	8	-	-
PC3. wear personal protective equipment while visiting the different departments during production. orexample mask in the washing section, glasses and mask in an assembly line, and gloves in the printing section, etc.	3	8	-	-
PC4. always wash sanitize your hands after a factory unit before touching any document, laptop, cell phone, etc.	3	8	-	-
PC5. undergo preventive health checkups at regular intervals.	3	8	-	-
PC6. take prompt treatment from the doctor in case of illness.	3	8	-	-
<i>Achieve work productivity while maintaining health</i>	9	25	-	-
PC7. follow SOPs for dealing with blisters; scratches; accidental fires or any other type of emergencies at work	3	8	-	-
PC8. ensure no productivity loss or absenteeism from work due to illness	3	8	-	-
PC9. ensure no long-term ill effect on personal health.	3	9	-	-
NOS Total	27	73	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N9903
NOS Name	Maintain Personal Hygiene
Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Ceramics), Handicrafts (Fashion Jewellery), Handicrafts (Stonecraft), Glassware, Metalware, Carpet, Handicrafts (Woodware)
Occupation	Generic
NSQF Level	4
Credits	1
Version	3.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N9904: Basic Business Management

Description

This OS unit focuses on business planning, internal strategy, timelines.

Scope

The scope covers the following :

- People management
- Product planning
- Procurement of raw materials
- Market interfacing
- Financial management
- Record keeping

Elements and Performance Criteria

People management

To be competent, the user/individual on the job must be able to:

- PC1.** arrange Interactive meetings of managers of sales and production teams and categorize the issues and feedbacks of both the teams
- PC2.** train the employees of his/her unit with the appropriate skills required to make marketrelevant and quality products
- PC3.** motivate the employees

Product planning

To be competent, the user/individual on the job must be able to:

- PC4.** compile a report based on old production reports
- PC5.** address the issues faced in previous productions and try to resolve them
- PC6.** gather and analyze the cues from the market
- PC7.** ascertain the customer preference
- PC8.** develop product range lines based on current market preference
- PC9.** develop product range lines that are unique and able to price high
- PC10.** price the products according to market trends
- PC11.** identify the competent marketing strategy for the product range

Procurement of raw materials

To be competent, the user/individual on the job must be able to:

- PC12.** list of the raw material s and prepare a B.O.M according to the product lines
- PC13.** ascertain the quantity and right price to procure the materials
- PC14.** identify the right locations/agents from where the raw materials can be procured
- PC15.** negotiate to get the best price
- PC16.** ensure quality materials are procured

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PC17. ensure the procured materials are stored in appropriate conditions

PC18. compile a record of price quotations, POs, and bills of procurement for future reference

Market interfacing

To be competent, the user/individual on the job must be able to:

PC19. maintain a healthy and professional relationship with vendor

PC20. the competitive market falls in order with the company policies of best price, quality, and delivery parameters

PC21. analyze the prevalent price for product lines

PC22. decide on the most effective means to access the market

PC23. plan for cost-effective transportation to the market

PC24. position the product according to market requirements

PC25. identify and address the expectations of customer

Financial management

To be competent, the user/individual on the job must be able to:

PC26. analyze and ascertain the cost of production

PC27. maintain the book of accounts related to the business

PC28. maintain export documents like a letter of credit, custom clearance

PC29. identify cost-effective means of running the business

Record keeping

To be competent, the user/individual on the job must be able to:

PC30. identify various aspects of business that require recording

PC31. design formats for recording

PC32. compile various records of all aspects of the business

PC33. maintain these records with periodic updations

PC34. maintain necessary documents as per local government and regulatory requirements

PC35. reframe the procurement strategy according to local scenarios like weather conditions, transport strikes, affected prices, etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. interpersonal skills and communication with a cross-section of stakeholders

KU2. understanding of basics accounting so that he/she can monitor the transactions between company and vendor

KU3. knowledge of banking basics to keep the export process flawless

KU4. understanding of costing principles to calculate the cost of production with all the overheads and the actual price

KU5. product and craft knowledge including material and tools requirements

KU6. gathering market intelligence.

KU7. various transportation means and implications on costing

KU8. various product lines that can be created depending on the sector of operation

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- KU9.** basic record-keeping techniques
- KU10.** basic laws, rules, regulations, etc. regarding business
- KU11.** vendor management and development
- KU12.** pricing techniques
- KU13.** business profitability assessment

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyse product range and keep abreast of market trends
- GS2.** document various aspects of the business
- GS3.** compile descriptions and details about investment, expenditures, and sales
- GS4.** interact with teams to work efficiently
- GS5.** communicate and manage vendors for cost-effective outsourcing and procurements
- GS6.** interface with fellow entrepreneurs to exchange ideas on the business
- GS7.** communicate with the customers for their feedback about the product
- GS8.** comprehend information shared by various stakeholders
- GS9.** plan and organize the product lines
- GS10.** fix the appropriate price.
- GS11.** hire employees with the appropriate skill set and experience
- GS12.** predict the profit margin to be achieved by the business
- GS13.** decide target segment of the market
- GS14.** schedule production cycles for better efficiency of resources
- GS15.** planning of production efficiency based on manpower and equipment available
- GS16.** schedule market visits for surveys and feedback
- GS17.** gather information on preference and taste of the customer
- GS18.** interact with various types of customers and understand the trends
- GS19.** analyze and solve conflicts and problems of the business.
- GS20.** ensure that the problems do not arise repeatedly.
- GS21.** anticipate various problems/challenges that can crop up
- GS22.** analyze the market for increasing sales
- GS23.** spot errors and any other disruptions and communicate with solutions

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>People management</i>	3	6	-	-
PC1. arrange Interactive meetings of managers of sales and production teams and categorize the issues and feedbacks of both the teams	1	2	-	-
PC2. train the employees of his/her unit with the appropriate skills required to make marketrelevant and quality products	1	2	-	-
PC3. motivate the employees	1	2	-	-
<i>Product planning</i>	8	16	-	-
PC4. compile a report based on old production reports	1	2	-	-
PC5. address the issues faced in previous productions and try to resolve them	1	2	-	-
PC6. gather and analyze the cues from the market	1	2	-	-
PC7. ascertain the customer preference	1	2	-	-
PC8. develop product range lines based on current market preference	1	2	-	-
PC9. develop product range lines that are unique and able to price high	1	2	-	-
PC10. price the products according to market trends	1	2	-	-
PC11. identify the competent marketing strategy for the product range	1	2	-	-
<i>Procurement of raw materials</i>	7	14	-	-
PC12. list of the raw material s and prepare a B.O.M according to the product lines	1	2	-	-
PC13. ascertain the quantity and right price to procure the materials	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the right locations/agents from where the raw materials can be procured	1	2	-	-
PC15. negotiate to get the best price	1	2	-	-
PC16. ensure quality materials are procured	1	2	-	-
PC17. ensure the procured materials are stored in appropriate conditions	1	2	-	-
PC18. compile a record of price quotations, POs, and bills of procurement for future reference	1	2	-	-
<i>Market interfacing</i>	7	14	-	-
PC19. maintain a healthy and professional relationship with vendor	1	2	-	-
PC20. the competitive market falls in order with the company policies of best price, quality, and delivery parameters	1	2	-	-
PC21. analyze the prevalent price for product lines	1	2	-	-
PC22. decide on the most effective means to access the market	1	2	-	-
PC23. plan for cost-effective transportation to the market	1	2	-	-
PC24. position the product according to market requirements	1	2	-	-
PC25. identify and address the expectations of customer	1	2	-	-
<i>Financial management</i>	4	8	-	-
PC26. analyze and ascertain the cost of production	1	2	-	-
PC27. maintain the book of accounts related to the business	1	2	-	-
PC28. maintain export documents like a letter of credit, custom clearance	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. identify cost-effective means of running the business	1	2	-	-
<i>Record keeping</i>	6	7	-	-
PC30. identify various aspects of business that require recording	1	2	-	-
PC31. design formats for recording	1	1	-	-
PC32. compile various records of all aspects of the business	1	1	-	-
PC33. maintain these records with periodic updations	1	1	-	-
PC34. maintain necessary documents as per local government and regulatory requirements	1	1	-	-
PC35. reframe the procurement strategy according to local scenarios like weather conditions, transport strikes, affected prices, etc.	1	1	-	-
NOS Total	35	65	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N9904
NOS Name	Basic Business Management
Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Ceramics), Handicrafts (Fashion Jewellery), Handicrafts (Stonecraft), Glassware, Metalware
Occupation	Generic
NSQF Level	4
Credits	1
Version	3.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 50

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N2801.Perform pre casting requirements	37	63	-	-	100	10
HCS/N2802.Casting of metal surface	33	67	-	-	100	20
HCS/N0603.Carry out the process involved in Sandcasting Techniques	30	70	-	-	100	10
HCS/N8505.Entrepreneurship skills for Metal casting operator	40	60	-	-	100	10
HCS/N9901.Coordinate with colleagues and work as a team	35	65	0	0	100	10
HCS/N9913.Maintain health, safety and security at workplace	30	70	-	-	100	10
HCS/N9903.Maintain Personal Hygiene	27	73	0	0	100	10
HCS/N9904.Basic Business Management	35	65	0	0	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	287	563	0	0	850	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.